

Regularized Algorithms for Online Optimization and Learning

CS245: Online Optimization and Learning

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2024 ACM A.M. Turing Award



Andrew Barto



Rich Sutton

Alan Turing: What we want is a machine that can learn from experience?

Answer: Reinforcement learning!

Keywords: Temporal difference learning, policy gradient methods,
RL book, AlphaGo, ChatGPT, RLHF

2024 ACM A.M. Turing Award



Richard S. Sutton

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TITLE	CITED BY	YEAR
Reinforcement learning: An introduction RS Sutton, AG Barto MIT press 1 (1), 9-11	79694	1998
Policy gradient methods for reinforcement learning with function approximation RS Sutton, D McAllester, S Singh, Y Mansour Advances in neural information processing systems 12	9288	1999
Learning to predict by the methods of temporal differences RS Sutton Machine learning 3, 9-44	8383	1988
Neuronlike adaptive elements that can solve difficult learning control problems AG Barto, RS Sutton, CW Anderson IEEE transactions on systems, man, and cybernetics 13 (5), 834-846	5301	1983
Between MDPs and semi-MDPs: A framework for temporal abstraction in reinforcement learning RS Sutton, D Precup, S Singh Artificial intelligence 112 (1-2), 181-211	4731	1999
Integrated architectures for learning, planning, and reacting based on approximating dynamic programming RS Sutton Proceedings of the International Conference on Machine Learning, 216-224	2326	1990

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TITLE	CITED BY	YEAR
Asynchronous Stochastic Approximation and Average-Reward Reinforcement Learning H Yu, Y Wan, RS Sutton arXiv preprint arXiv:2409.03915		2024
On Convergence of Average-Reward Q-Learning in Weakly Communicating Markov Decision Processes Y Wan, H Yu, RS Sutton arXiv preprint arXiv:2408.16262	1	2024
Loss of plasticity in deep continual learning S Dohare, JF Hernandez-Garcia, Q Lan, P Rahman, AR Mahmood, ... Nature 632 (8026), 768-774	60	2024
An Idiosyncrasy of Time-discretization in Reinforcement Learning K De Asis, RS Sutton arXiv preprint arXiv:2406.14951	1	2024
Off-Policy Prediction Learning: An Empirical Study of Online Algorithms S Ghiassian, B Rafiee, RS Sutton IEEE Transactions on Neural Networks and Learning Systems	1	2024
Reward centering A Naik, Y Wan, M Tomar, RS Sutton arXiv preprint arXiv:2405.09999	9	2024

Online Gradient Descent (OGD)

Initialization: $x_1 \in \mathcal{K}$ and $\{\eta_t\}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \Pi_{\mathcal{K}}(x_t - \eta_t \nabla f_t(x_t))$.
-

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The intuition of OGD is to solve “trust region optimization”:

$$\begin{aligned} \min_{x \in \mathcal{K}} \quad & f_t(x_t) + \langle x - x_t, \nabla f_t(x_t) \rangle \\ \text{s.t.} \quad & \|x - x_t\| \leq \delta. \end{aligned}$$

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-

The intuition of OGD is to minimize the first order approximation + regularization with ℓ_2 norm:

$$\hat{f}_{t+1}(x) = f_t(x_t) + \langle x - x_t, \nabla f_t(x_t) \rangle + \frac{1}{2\eta_t} \|x - x_t\|^2.$$

which is equivalent to

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{2\eta_t} \|x - x_t\|^2.$$

Bregman Divergence

Definition 1 (Bregman Divergence)

Let $\psi : X \rightarrow R$ be strongly convex and continuously differentiable function. The Bregman divergence w.r.t. ψ is B_ψ is defined as

$$B_\psi(x; y) = \psi(x) - \psi(y) - \langle x - y, \nabla \psi(y) \rangle.$$

If ψ is twice differentiable, and by Taylor theorem

$$B_\psi(x; y) = \langle x - y, \nabla^2 \psi(z)(x - y) \rangle,$$

where z is a point between x and y .

Recall $\psi(\cdot)$ is α -strongly convex, we have a global property

$$B_\psi(x; y) \geq \frac{\alpha}{2} \|x - y\|^2.$$

Bregman Divergence - properties

The properties of Bregman divergence:

- Non-negative

$$B_{\psi}(x; y) \geq 0.$$

- “Non”-symmetric

$$B_{\psi}(x; y) \neq B_{\psi}(y; x).$$

- Three points identity:

$$B_{\psi}(z; x) + B_{\psi}(x; y) - B_{\psi}(z; y) = \langle \nabla \psi(y) - \nabla \psi(x), z - x \rangle.$$

Online Mirrored Descent

Online gradient descent is

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{2\eta_t} \|x - x_t\|^2.$$

Online Mirrored Descent

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$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{2\eta_t} \|x - x_t\|^2.$$

Just change the “distance” metric to Bregman divergence w.r.t ψ , and we have

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta_t} B_\psi(x; x_t).$$

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If $\mathcal{K}$ is $\mathbb{R}^d$, let $\psi(x) = \frac{1}{2}\|x\|^2$ gives us online gradient descent algorithm.

If $\mathcal{K}$ is a probability simplex, let $\psi(x) = \sum_{i=1}^d x_i \log x_i$ gives us **any algorithm?**

Online Mirrored Descent (OMD)

Initialization: $x_1 \in \mathcal{K}$ and $\{\eta_t\}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta_t} B_\psi(x; x_t)$.
-

An alternative update is

$$y_{t+1} = \arg \min_{x \in \mathbb{R}^d} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta_t} B_\psi(x; x_t)$$

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} B_\psi(x; y_{t+1})$$

Online Mirrored Descent - Regret

Recall the regret of online gradient descent is $O(\sqrt{T})$. How about the regret of online mirrored descent?

Theorem 2

Let ψ be α -strongly convex function. Consider a fixed learning rate $\eta_t = \eta$. Online mirrored descent algorithm achieves

$$\text{Regret}(T) \leq \frac{B_\psi(x^*, x_1)}{\eta} + \frac{1}{2\alpha} \sum_{t=1}^T \eta \|\nabla f_t(x_t)\|^2.$$

OMD achieves $O(\sqrt{T})$ regret if:

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OMD achieves $O(\sqrt{T})$ regret if:

- The feasible set and gradients are bounded.

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OMD achieves $O(\sqrt{T})$ regret if:

- The feasible set and gradients are bounded.
- Learning rate is fixed with $O(1/\sqrt{T})$.

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OMD achieves $O(\sqrt{T})$ regret if:

- The feasible set and gradients are bounded.
- Learning rate is fixed with $O(1/\sqrt{T})$.
- Time varying learning rate $O(1/\sqrt{t})$ or adaptive learning rate also work (verify by yourself).

Online Mirrored Descent - Proof

We use a “potential/Lyapunov drift” style of analysis: define

$$\begin{aligned}\phi_t &= B_\psi(x^*; x_t) \\ &= \psi(x^*) - \psi(x_t) - \langle x^* - x_t, \nabla \psi(x_t) \rangle,\end{aligned}$$

and study the drift

$$\begin{aligned}\phi_{t+1} - \phi_t &= B_\psi(x^*; x_{t+1}) - B_\psi(x^*; x_t) \\ &= -B_\psi(x_{t+1}; x_t) + \langle \nabla \psi(x_t) - \nabla \psi(x_{t+1}), x^* - x_{t+1} \rangle\end{aligned}$$

Define $F_t(x) = \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x; x_t)$

We have $x_{t+1} = \arg \min F_t(x)$, which implies

$$\langle x^* - x_{t+1}, \nabla F_t(x_{t+1}) \rangle \geq 0 \quad \text{First-order optimality}$$

Online Mirrored Descent - Proof

Substitute $\nabla f_t(x_t)$ and we have

$$\langle x^* - x_{t+1}, \nabla f_t(x_t) + \frac{1}{\eta} \nabla \psi(x_{t+1}) - \frac{1}{\eta} \nabla \psi(x_t) \rangle \geq 0$$

$$\Rightarrow \langle \nabla \psi(x_t) - \nabla \psi(x_{t+1}), x^* - x_{t+1} \rangle \leq \eta \langle x^* - x_{t+1}, \nabla f_t(x_t) \rangle$$

we finally have

$$\begin{aligned} & \langle x_{t+1}, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x_{t+1}; x_t) \\ & \leq \langle x^*, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1}) \end{aligned}$$

which is exactly the pushback lemma in the following alternative proof. the remaining proof can be found in the alternative proof next.

Online Mirrored Descent - An Alternative Proof

We have the following lemma that make our analysis simple¹

Lemma 3 (A pushback lemma)

Suppose x_{t+1} minimizes the function $F(x)$ such that

$$F(x) := \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x; x_t),$$

For any x , we have

$$F(x_{t+1}) \leq F(x) - \frac{1}{\eta} B(x; x_{t+1}).$$

¹X. Wei, et al. Online Primal-Dual Mirror Descent under Stochastic Constraints. Sigmetrics 2020.

Online Mirrored Descent - An Alternative Proof

By Lemma 3, we have for any x such that

$$\begin{aligned} & \langle x_{t+1}, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x_{t+1}; x_t) \\ & \leq \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x; x_t) - \frac{1}{\eta} B(x; x_{t+1}) \end{aligned}$$

Let $x = x^*$, then we have

$$\begin{aligned} & \langle x_{t+1} - x_t, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x_{t+1}; x_t) \\ & \leq \langle x^* - x_t, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1}) \\ & \leq f_t(x^*) - f_t(x_t) + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1}) \end{aligned}$$

$$\Rightarrow f(x_t) - f(x^*)$$

$$\leq -\langle x_{t+1} - x_t, \nabla f_t(x_t) \rangle - \frac{1}{\eta} B(x_{t+1}; x_t) + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1})$$

$$\leq -\langle x_{t+1} - x_t, \nabla f_t(x_t) \rangle - \frac{2}{2\eta} \|x_{t+1} - x_t\|^2 + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1})$$

$$= -\langle x_{t+1} - x_t, \nabla f_t(x_t) \rangle - \frac{2}{2\eta} \|x_{t+1} - x_t\|^2 - \frac{\eta}{2\alpha} \|\nabla f_t(x_t)\|^2 + \frac{\eta}{2\alpha} \|\nabla f_t(x_t)\|^2 + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1})$$

$$\leq \frac{\eta}{2\alpha} \|\nabla f_t(x_t)\|^2 + \frac{1}{\eta} B(x^*; x_t) - \frac{1}{\eta} B(x^*; x_{t+1})$$

therefore, we have

$$\sum_{t=1}^T f_t(x_t) - f_t(x^*) \leq \frac{1}{\eta} B(x^*; x_1) + \sum_{t=1}^T \frac{\eta}{2\alpha} \|\nabla f_t(x_t)\|^2$$

which proves the Theorem.

Why is called Mirrored descent?

Definition 4 (Fenchel Conjugate)

The Fenchel conjugate of a function f is

$$f^*(y) := \sup_{x \in \mathcal{K}} \langle y, x \rangle - f(x).$$

Theorem 5

The update of online mirrored descent

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta_t} B_\psi(x; x_t)$$

is equivalent to

$$x_{t+1} = \nabla \psi_{\mathcal{K}}^*(\nabla \psi_{\mathcal{K}}(x_t) - \eta_t \nabla f_t(x_t)).$$

Let's consider the case of $\psi(x) = \frac{1}{2} \|x\|^2$, can we reduce it to online gradient descent?

Theorem 5 – Proof

By definition of online mirror descent, we have

$$\begin{aligned}x_{t+1} &= \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta_t} B_\psi(x; x_t) \\&= \arg \min_{x \in \mathcal{K}} \eta_t \langle x, \nabla f_t(x_t) \rangle + B_\psi(x; x_t) \\&= \arg \min_{x \in \mathcal{K}} \eta_t \langle x, \nabla f_t(x_t) \rangle + \psi(x) - \langle x, \nabla \psi(x_t) \rangle \\&= \arg \min_{x \in \mathcal{K}} \langle x, \eta_t \nabla f_t(x_t) - \nabla \psi(x_t) \rangle + \psi(x) \\&= \arg \max_{x \in \mathcal{K}} \langle x, \nabla \psi(x_t) - \eta_t \nabla f_t(x_t) \rangle - \psi(x)\end{aligned}$$

Let's define $y = \nabla \psi(x_t) - \eta_t \nabla f_t(x_t)$, and we have

$$x_{t+1} = \arg \max_{x \in \mathcal{K}} \langle y, x \rangle - \psi(x).$$

Theorem 5 – Proof

Let's first consider $\mathcal{K} = \mathbb{R}^d$. Note x_{t+1} is maximizing

$$\langle x, y \rangle - \psi(x),$$

we have

$$\begin{aligned}\nabla \psi^*(y) &= \frac{\partial (\max_x \langle x, y \rangle - \psi(x))}{\partial y}, \\ &= \frac{\partial (\langle x_{t+1}, y \rangle - \psi(x_{t+1}))}{\partial y} \\ &= x_{t+1},\end{aligned}$$

which means

$$x_{t+1} = \nabla \psi^*(y) = \nabla \psi^*(\nabla \psi(x_t) - \eta_t \nabla f_t(x_t)).$$

We are done. Please verify the case of the general $\mathcal{K}$.

Why is called Mirrored descent?

Let's understand online mirrored descent ($\mathcal{K} = \mathbb{R}^d$)

$$x_{t+1} = \nabla\psi^*(\nabla\psi(x_t) - \eta_t \nabla f_t(x_t))$$

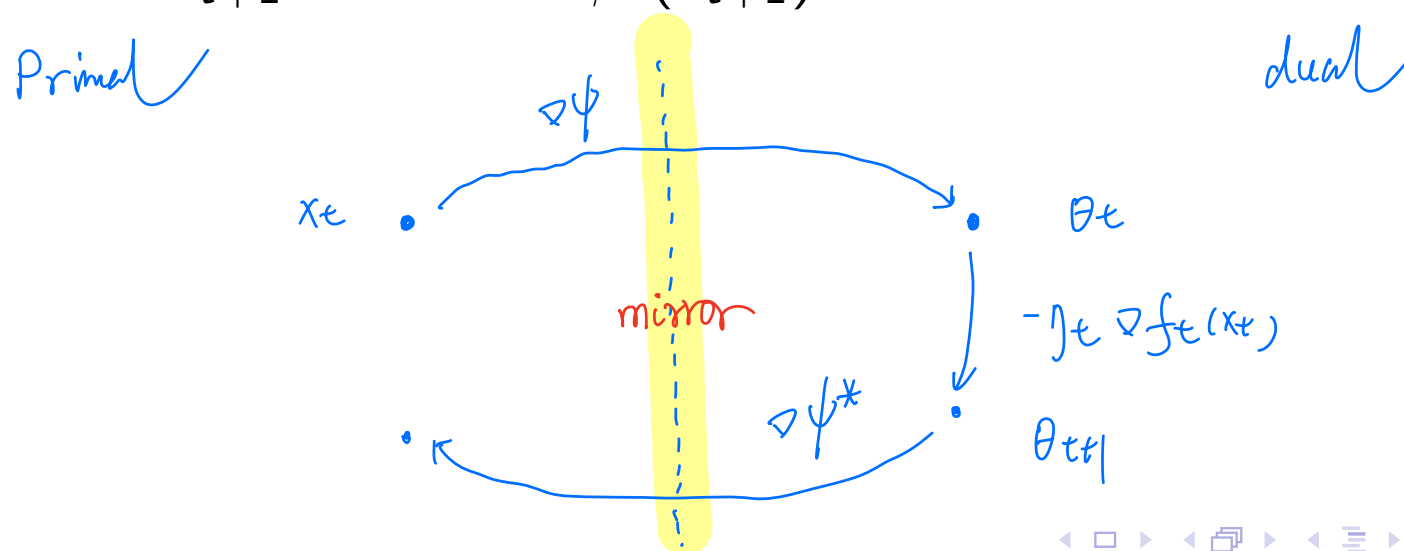
in three steps:

- Mirror x_t from primal space to dual $\theta_t = \nabla\psi(x_t)$.

- Take gradient descent in dual space

$$\theta_{t+1} = \theta_t - \eta_t \nabla f_t(x_t).$$

- Mirror θ_{t+1} back to $\nabla\psi^*(\theta_{t+1})$.



Review of Expert problem

Expert problem:

Initialization: N experts/models.

For each day $t = 1, \dots, T$:

- **Learner:** Obtain predictions from N experts/models and sample an expert i from a probability simplex x_t .
 - **Environment:** Observe the loss of each model $\ell_t \in [0, 1]^N$.
-

Objective: Find the best expert in hindsight, which is equivalent to minimize regret:

$$\mathcal{R}(T) := \mathbb{E} \left[\sum_{t=1}^T \ell_t(i) - \sum_{t=1}^T \ell_t(i^*) \right] = \sum_{t=1}^T \langle x_t, \ell_t \rangle - \sum_{t=1}^T \langle x^*, \ell_t \rangle$$

Expert problem: Hedge

Hedge - “weighted” version:

Initialization: $w_1(i) = 1, \forall i \in [N]$.

For each day $t = 1, \dots, T$:

- **Learner:** Sample an expert i : $p_t(i) = w_t(i) / \sum_i w_t(i)$.
 - **Environment:** Observe the error $\ell_t \in [0, 1]^N$.
 - **Update:** $w_{t+1} = w_t \cdot e^{-\eta \ell_t(i)}, \forall i \in [N]$.
-

Hedge - “prob” version:

Initialization: $x_1 = [1/d, \dots, 1/d]$ and η .

For each day $t = 1, \dots, T$:

- **Learner:** Sample an expert i according to x_t .
 - **Environment:** Observe the error $\ell_t \in [0, 1]^N$.
 - **Update:** $x_{t+1,i} = x_{t,i} e^{-\eta \ell_t(i)} / \sum_{i=1}^d x_{t,i} e^{-\eta \ell_t(i)}, \forall i \in [N]$.
-

Exponentiated Gradient – Hedge

Exponentiated Gradient:

Initialization: $x_1 = [1/d, \dots, 1/d]$ and η .

For each day $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the loss $f_t(\cdot)$.
 - **Update:** $x_{t+1,i} = x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)} / \sum_{i=1}^d x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)}$.
-

How it is related to Hedge - “prob” version?

- No sampling operator from x_t .
- The loss is $f_t(x_t) = \langle x_t, \ell_t \rangle$.
- Regret is equivalent to the “expected” regret of Hedge!

Online Mirrored Descent:

Initialization: $x_1 = [1/d, \dots, 1/d]$ and η .

For each day $t = 1, \dots, T$:

- **Learner:** submit x_t .
 - **Environment:** Observe the loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{\mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B_{\psi}(x; x_t)$.
-

Since x in the prob simplex, can we try $\psi(x) = \sum_{i=1}^d x_i \log x_i$ in the Bregman divergence and show x_{t+1} is equivalent to that in Exponentiated Gradient?

Exponentiated Gradient – Online Mirrored Descent

Online Mirrored Descent:

Initialization: $x_1 = [1/d, \dots, 1/d]$ and η .

For each day $t = 1, \dots, T$:

- **Learner:** submit x_t .
 - **Environment:** Observe the loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{\mathcal{K}} \langle x, \nabla f_t(x_t) \rangle + \frac{1}{\eta} B_{\psi}(x; x_t)$.
-

Since x in the prob simplex, can we try $\psi(x) = \sum_{i=1}^d x_i \log x_i$ in the Bregman divergence and show x_{t+1} is equivalent to that in the Exponentiated Gradient:

$$x_{t+1,i} = \frac{x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)}}{\sum_{j=1}^d x_{t,j} e^{-\eta \nabla f_{t,j}(x_t)}}.$$

Exponentiated Gradient as Online Mirrored Descent

The update of Bragman divergence

$$\begin{aligned} \min_{x \in \mathcal{K}} \quad & \eta \langle x, \nabla f_t(x_t) \rangle + \sum_{i=1}^d x_i \log \frac{x_i}{x_{t,i}} \\ \text{s.t.} \quad & \sum_{i=1}^d x_i = 1, \quad x_i \geq 0. \end{aligned}$$

Let's consider (partial) Lagrangian function:

$$L(x, \lambda) = \eta \langle x, \nabla f_t(x_t) \rangle + \sum_{i=1}^d x_i \log \frac{x_i}{x_{t,i}} + \lambda \left(1 - \sum_{i=1}^d x_i\right)$$

$$\frac{\partial L(x, \lambda)}{\partial x_i} = \eta \nabla f_{t,i}(x_t) + \log \frac{x_i}{x_{t,i}} + 1 - \lambda = 0 \quad (i)$$

$$\frac{\partial L(x, \lambda)}{\partial \lambda} = 1 - \sum_{i=1}^d x_i = 0 \quad (ii)$$

Exponentiated Gradient as Online Mirrored Descent

From (i), we have

$$\log x_i = \log x_{t,i} - \eta \nabla f_{t,i}(x_t) + \lambda - 1$$

$$\Rightarrow x_i = x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)} e^{\lambda - 1}$$

Combine (ii), we normalize the x_i such that

$$x_i = \frac{x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)}}{\sum_{i=1}^d x_{t,i} e^{-\eta \nabla f_{t,i}(x_t)}}$$

Hedge as Online Mirrored Descent

Hedge as Online Mirrored Descent:

Initialization: $x_1 = [1/d, \dots, 1/d]$ and η_t .

For each day $t = 1, \dots, T$:

- **Learner:** Sample an expert i from x_t .
 - **Environment:** Observe the error $\ell_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{\mathcal{K}} \langle x, \ell_t \rangle + \frac{1}{\eta} B_{\psi}(x; x_t)$.
-

Hedge $\longrightarrow$ Exponentiated Gradient $\longrightarrow$ OMD!

OMD is a strong and general framework to design online algorithms!

Theorem 6 (Restate Theorem 2)

Let ψ be α -strongly convex function in B_ψ . Let fixed learning rate $\eta_t = \eta$. Online mirrored descent algorithm achieves

$$\text{Regret}(T) \leq \frac{B_\psi(x^*, x_1)}{\eta} + \frac{\eta}{2\alpha} \sum_{t=1}^T \|\nabla f_t(x_t)\|^2.$$

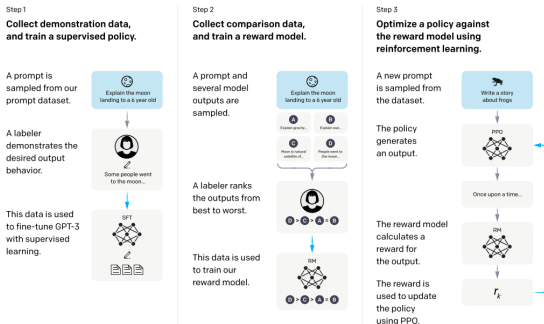
In Hedge, we have

- $\psi(x) = \sum_{i=1}^d x_i \log x_i$ is 1-strongly convex,
- $B_\psi(x^*, x_1) = \sum_{i=1}^d x_i^* \log \frac{x_i^*}{x_{1,i}} \leq \log N$,

which implies the regret of Hedge is

$$\text{Regret}(T) = O(\sqrt{T \log N}).$$

LLM Finetuning



“Reinforcement learning from human feedback” for LLM finetuning²:

- Learn a base/ref policy from supervise learning
- Learn reward models via human/AI feedback
- Policy optimization based on reward models

²Long Ouyang, etc., “Training language models to follow instructions with human feedback”. NeurIPS, 2022.

RLHF Finetuning as (Online) Mirrored Descent

RLHF³ as (Online) Mirrored Descent:

Initialization: A pretrained reward model r and policy π_{ref} .

FineTuning: Find a policy such that

$$\pi_{\theta}^* = \arg \max \mathbb{E}_{c \sim \mathcal{D}, a \sim \pi_{\theta}(a|c)} [r(c, a)] - \frac{1}{\eta} \text{KL}(\pi_{\theta}; \pi_{ref}).$$

³OpenAI. Training language models to follow instructions with human feedback. NeurIPS 2022.

Online Learning with Prediction

Consider a linear function

$$f_t(x) = \langle \ell_t, x \rangle.$$

Online Learning with Prediction

Initialization: $x_1 \in \mathcal{K}$ and $\{\eta_t\}$.

For $t = 1, \dots, T$:

- **Learner:** Given a prediction $\hat{\ell}_t$ and submit x_t .
 - **Environment:** Observe the cost ℓ_t .
-

Online Learning with Prediction

Consider a linear function

$$f_t(x) = \langle \ell_t, x \rangle.$$

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-



predict weather tomorrow in shanghai



As an AI language model, I cannot predict the weather with certainty. However, based on historical weather data and current trends, I can suggest that the weather in Shanghai tomorrow is likely to be partly cloudy with a chance of scattered showers. The temperature is expected to range from around 10 to 18 degrees Celsius, with a moderate breeze from the east-northeast. However, I would recommend checking the latest weather forecast for Shanghai to get the most accurate information.



list three top papers on this topic with just titles



Sure, here are three top papers on online learning and optimization with just their titles:



1. "Online Learning and Online Convex Optimization" by Elad Hazan.
2. "Adaptive Subgradient Methods for Online Learning and Stochastic Optimization" by John Duchi, Elad Hazan, and Yoram Singer.
3. "Follow-the-Regularized-Leader and Mirror Descent: Equivalence Theorems and L1 Regularization" by Peter Bartlett, Elad Hazan, and Alexander Rakhlin.

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-

 **You**
predict weather tomorrow in shanghai

 **ChatGPT**
I'm sorry for any confusion, but as an AI, I don't have access to real-time data or the ability to predict the weather. For accurate and up-to-date weather forecasts for Shanghai or any other location, I recommend checking a reliable weather forecasting website or app, or tuning in to a local weather report. These sources typically provide the most accurate and timely information about weather conditions.

 **You**
list three top papers on this topic with just titles

 **ChatGPT**

1. "Regret Analysis of Stochastic and Nonstochastic Multi-armed Bandit Problems" by Peter Auer, Nicolo Cesa-Bianchi, and Paul Fischer.
2. "Follow the Regularized Leader and Mirror Descent: Equivalence Theorems and L1 Regularization" by Elad Hazan, Amit Agarwal, and Satyen Kale.
3. "Online Learning: A Survey" by Shai Shalev-Shwartz.

Online Learning with Prediction

Consider a linear function

$$f_t(x) = \langle \ell_t, x \rangle.$$

Online Learning with Prediction

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-

How to utilize the prediction to improve the online learning algorithms?

- For perfect predictions $\hat{\ell}_t = \ell_t$, the regret is smaller than $O(\sqrt{T})$?
- For bad predictions, the regret should not be worse than $O(\sqrt{T})$!

Online Learning with Prediction

Initialization: $x_1 \in \mathcal{K}$ and $\{\eta\}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the cost ℓ_t .
 - **Prediction:** The cost $\hat{\ell}_{t+1}$.
 - **Update:** $x_{t+1} = \text{Alg}(x_1, \dots, x_t, \ell_1, \dots, \ell_t, \hat{\ell}_{t+1})$.
-

$\text{Alg}(x_1, \dots, x_t, \ell_1, \dots, \ell_t, \hat{\ell}_{t+1})$ could be $\text{Alg}(x_t, \ell_t, \hat{\ell}_{t+1})$ like online gradient/mirrored descent:

$$x_{t+1} = \arg \min_{x \in \mathbb{R}^d} \langle x, \ell_t \rangle + \frac{1}{\eta} B_\psi(x; x_t)$$

How to incorporate the prediction $\hat{\ell}_{t+1}$?

Online Learning with Prediction

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Online gradient/mirrored descent:

$$y_{t+1} = \arg \min_{y \in \mathbb{R}^d} \langle y, \ell_t \rangle + \frac{1}{\eta} B_\psi(y; y_t)$$

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-

Online gradient/mirrored descent **with prediction**:

$$y_{t+1} = \arg \min_{y \in \mathbb{R}^d} \langle y, \ell_t \rangle + \frac{1}{\eta} B_\psi(y; y_t)$$

$$x_{t+1} = \arg \min_{x \in \mathbb{R}^d} \langle x, \hat{\ell}_{t+1} \rangle + \frac{1}{\eta} B_\psi(x; y_{t+1})$$

Online Mirrored Descent with Prediction

Initialization: $x_1 \in \mathcal{K}$ and $\{\eta\}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
- **Environment:** Observe the loss ℓ_t .
- **Prediction:** The cost $\hat{\ell}_{t+1}$.
- **Update:** $y_{t+1} = \arg \min_{y \in \mathbb{R}^d} \langle y, \ell_t \rangle + \frac{1}{\eta} B_\psi(y; y_t)$
 $x_{t+1} = \arg \min_{x \in \mathbb{R}^d} \langle x, \hat{\ell}_{t+1} \rangle + \frac{1}{\eta} B_\psi(x; y_{t+1})$

Intuition:

- Online mirrored descent guarantees “not too bad” even with unreliable predictions.
- Decrease the cost further if $\hat{\ell}_{t+1}$ is reliable.

Online Mirrored Descent with Prediction – Regret

The regret of OMD with prediction⁴ is as follows.

Theorem 7

Let ψ be 1-strongly convex function in B_ψ . Let fixed learning rate $\eta_t = \eta$. Given a prediction sequence of $\{\hat{\ell}_t\}$, online mirrored descent achieves

$$\text{Regret}(T) \leq \frac{B(x^*, x_1)}{\eta} + \frac{\eta}{2} \sum_{t=1}^T \|\hat{\ell}_t - \ell_t\|^2.$$

“Almost” the best of two worlds:

⁴Alexander Rakhlin and Karthik Sridharan. Online learning with predictable sequences. COLT, 2013

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- If the predictions are “perfect”, the regret is constant!

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- If the predictions are “bad”, the regret can be $O(\sqrt{T})$.

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“Almost” the best of two worlds:

- If the predictions are “perfect”, the regret is constant!
- If the predictions are “bad”, the regret can be $O(\sqrt{T})$.
- If the predictions are “good”, the regret can be $o(\sqrt{T})$.

⁴Alexander Rakhlin and Karthik Sridharan. Online learning with predictable sequences. COLT, 2013

Review: OMD without Prediction – Proof

According to the pushback lemma, suppose x_{t+1} minimizes the function $F(x)$ such that

$$F(x) := \langle x, \ell_t \rangle + \frac{1}{\eta} B(x; x_t).$$

For any x , we have

$$F(x_{t+1}) \leq F(x) - \frac{1}{\eta} B(x; x_{t+1}).$$

Therefore, we have

$$\eta \langle x_{t+1}, \ell_t \rangle + B(x_{t+1}; x_t) \leq \eta \langle x^*, \ell_t \rangle + B(x^*; x_t) - B(x^*; x_{t+1}).$$

which implies

$$\eta \langle x_t - x^*, \ell_t \rangle + \eta \langle x_{t+1} - x_t, \ell_t \rangle + B(x_{t+1}; x_t) \leq B(x^*; x_t) - B(x^*; x_{t+1}).$$

OMD with Prediction – Proof

Step one:

$$y_{t+1} = \arg \min_{y \in \mathbb{R}^d} \langle y, \ell_t \rangle + \frac{1}{\eta} B_\psi(y; y_t).$$

By pushback lemma, we have

$$\eta \langle y_{t+1}, \ell_t \rangle + B(y_{t+1}; y_t) \leq \eta \langle x^*, \ell_t \rangle + B(x^*; y_t) - B(x^*; y_{t+1}).$$

Step two:

$$x_t = \arg \min_{x \in \mathbb{R}^d} \langle x, \hat{\ell}_t \rangle + \frac{1}{\eta} B_\psi(x; y_t).$$

By pushback lemma, we have **for any x**

$$\eta \langle x_t, \hat{\ell}_t \rangle + B(x_t; y_t) \leq \eta \langle x, \hat{\ell}_t \rangle + B(x; y_t) - B(x; x_t).$$

Online Mirrored Descent with Prediction – Proof

From (i), we have

$$\begin{aligned} & \eta \langle x_t - x^*, l_t \rangle + \eta \langle y_{t+1} - x_t, l_t \rangle \\ & \leq -B(y_{t+1}; y_t) + B(x^*; y_t) - B(x^*; y_{t+1}) \end{aligned}$$

From (ii), we have

$$\begin{aligned} & \eta \langle x_t - y_{t+1}, \hat{c}_t \rangle \\ & \leq B(y_{t+1}; y_t) - B(y_{t+1}; x_t) \end{aligned}$$

Sum two inequality above, we have

$$\eta \langle x_t - x^*, l_t \rangle + \eta \langle y_{t+1} - x_t, l_t - \hat{l}_t \rangle$$

$$\leq B(x^*; y_t) - B(x^*; y_{t+1}) - B(y_{t+1}; x_t)$$

$$\leq B(x^*; y_t) - B(x^*; y_{t+1}) - \frac{1}{2} \|y_{t+1} - x_t\|^2$$

We have got this type of inequality a few times.

please complete the remaining proof.

Why Online Gradient/Mirrored Descent?

Online Learning Algorithm

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the cost ℓ_t .
 - **Update:** $x_{t+1} = \text{Alg}(x_1, \dots, x_t, \ell_1, \dots, \ell_t)$.
-

We design online learning algorithms to achieve small regret:

- Online gradient/mirrored descent is based on the current x_t and ℓ_t as

$$\text{Alg}(x_t, \ell_t).$$

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We design online learning algorithms to achieve small regret:

- Online gradient/mirrored descent is based on the current x_t and ℓ_t as

$$\text{Alg}(x_t, \ell_t).$$

- Can we use all information to design online algorithms?

$$x_{t+1} = \text{Alg}(x_1, \dots, x_t, \ell_1, \dots, \ell_t).$$

Follow-The-Leader (FTL) Algorithm

Follow-The-Leader (FTL) Algorithm

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x)$.
-

Intuition of Follow-The-Leader (FTL) algorithm:

- A batch/offline learning problem to use all history info.
- Minimize the “regret” for the next round

$$\sum_{s=1}^t f_s(x_{t+1}) \leq \sum_{s=1}^t f_s(x^*).$$

Follow-The-Leader (FTL) Algorithm – Regret

Theorem 8

Under Follow-The-Leader algorithm, we have the sequence of actions $\{x_t\}$ which satisfies

$$\begin{aligned}\mathcal{R}(T) &:= \sum_{t=1}^T f_t(x_t) - \min_{x \in \mathcal{K}} \sum_{t=1}^T f_t(x) \\ &\leq \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x_{t+1}).\end{aligned}$$

Intuitively, we have a small regret if it is “stable”:

x_t is close to x_{t+1} .

f_t is convex: $f_t(x_t) - f_t(x_{t+1}) \leq \langle \nabla f_t(x_t), x_t - x_{t+1} \rangle$

We want $\|x_t - x_{t+1}\| = O(\frac{1}{\sqrt{T}})$ to recover $O(\sqrt{T})$ regret!

Follow-The-Leader (FTL) Algorithm – Proof

We prove it by (informal) induction, please finish the formal proof!

$$\text{FTL} : x_{t+1} = \arg \min_x \sum_{s=1}^t f_s(x_s).$$

$$t=2 \quad f_1(x_2) \leq f_1(x^*)$$

$$t=3 \quad \begin{array}{c} \text{+} \\ f_1(x_3) + f_2(x_3) \end{array} \leq f_1(x^*) + f_2(x^*)$$

$$t=4 \quad \begin{array}{c} \text{+} \\ f_1(x_4) + f_2(x_4) + f_3(x_4) \end{array} \leq f_1(x^*) + f_2(x^*) + f_3(x^*)$$

⋮

$$t=T \quad f_1(x_2) + f_2(x_3) + \dots + f_T(x_{T+1}) \leq \sum_{t=1}^T f_t(x^*)$$

Follow-The-Leader (FTL) Algorithm – Caveat

Follow-The-Leader (FTL) Algorithm

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

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-

Let's consider a counter example as follows

$$\mathcal{K} = [-1, 1],$$
$$\{f_1, f_2, f_3, f_4, f_5, \dots, f_T\} = \{0.5x, -x, x, -x, x, \dots, x\}.$$

What is the regret of FTL algorithms?

Follow-The-Leader (FTL) Algorithm – Caveat

t	1	2	3	4		T
$f_t(x)$	$0.5x$	$-x$	x	$-x$		x
$\sum_{s=1}^t f_s(x)$	$0.5x$	$-0.5x$	$0.5x$	$-0.5x$		$0.5x$
x_t	0	-1	+1	-1	not stable!	
$f_t(x_t)$	0	+1	+1	+1		

Let's compute the regret:

$$\sum_{t=1}^T f_t(x_t) = T - 1$$

$$R(T) = T - 0.5$$

$$\min_x \sum_{t=1}^T f_t(x) = \min_x 0.5x = -0.5$$

FTRL Algorithm – Regret

Let's consider the linear costs and the quadratic regularizer:

$$f_t(x) = \langle \ell_t, x \rangle, \forall t, \quad R(x) = \frac{1}{2\eta} \|x\|^2.$$

Theorem 9 (linear losses and quadratic regularizer)

Assume $\|x - y\| \leq D, \forall x, y \in \mathcal{K}$ $\|\nabla f_t(x)\| \leq G, \forall x \in \mathcal{K}$.

Under Follow-The-Regularized-Leader algorithm, we have the sequence of actions $\{x_t\}$ which satisfies

$$\mathcal{R}(T) \leq DG\sqrt{2T}.$$

We recover the good result of $O(\sqrt{T})$, which is similar as online gradient descent.

We can also get similar result for a convex loss and other types of regularizer.

FTRL Algorithm – Proof

Main idea : $FTRL \xRightarrow{\text{reduce to}} FTL$

$FTRL$ is FTL such that

$$\begin{array}{ccccccc} f_0(x) & f_1(x) & f_2(x) & \dots & f_T(x) \\ R(x) & f_1(x) & f_2(x) & \dots & f_T(x) \end{array}$$

$$\begin{aligned} R^{FTL}(T) &= \sum_{t=0}^T f_t(x_t) - \sum_{t=0}^T f_t(x^*) \\ &\leq \sum_{t=0}^T f_t(x_t) - \sum_{t=0}^T f_t(x_{t+1}) \end{aligned}$$

$$\begin{aligned}
R^{FTRL}(T) &= \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x^*) \\
&\leq f_0(x^*) - f_0(x_0) + \sum_{t=1}^T f_t(x_t) - \sum_{t=1}^T f_t(x_{t+1}) \\
&\quad + f_0(x_0) - f_0(x_1) \\
&= R(x^*) - R(x_1) + \sum_{t=1}^T f_t(x_t) - f_t(x_{t+1}) \\
&\leq \frac{1}{2\eta} \|x^*\|^2 + \sum_{t=1}^T f_t(x_t) - f_t(x_{t+1})
\end{aligned}$$

Linear cost $f_t(x) = \langle l_t, x \rangle$ and we have

$$\sum_{t=1}^T f_t(x_t) - f_t(x_{t+1}) = \sum_{t=1}^T \langle l_t, x_t - x_{t+1} \rangle$$

By FTRL, we have

$$\begin{aligned}
x_{t+1} &= \arg \min_x \sum_{s=1}^t f_s(x) + R(x) \\
&= \arg \min_x \left\langle \sum_{s=1}^t l_s, x \right\rangle + \frac{1}{2\eta} \|x\|^2 \triangleq F_{t+1}(x)
\end{aligned}$$

$$\frac{\partial F_{t+1}(x)}{\partial x} = \sum_{s=1}^t l_s + \frac{1}{\eta} x = 0 \quad \Rightarrow \quad x_{t+1} = -\eta \sum_{s=1}^t l_s$$

Now $x_t - x_{t+1} = \eta l_t$, we finally have

$$\Rightarrow \frac{1}{2\eta} \|x_t - x_{t+1}\|^2 \leq f_t(x_t) - f_t(x_{t+1})$$

$$\leq G \|x_t - x_{t+1}\|$$

We have $\|x_t - x_{t+1}\| \leq 2\eta G$ to recover the same regret

$$R^{\text{FTRL}}(T) \leq \frac{1}{2\eta} D^2 + \sum_{t=1}^T \eta G^2$$

Reduction: from Linear to Convex

$$R^{\text{FTRL-Linear}} = O(\sqrt{T})$$

$$R^{\text{FTRL-convex}}$$

$$= \sum_{t=1}^T f_t(x_t) - f_t(x^*) \leq \sum_{t=1}^T \langle x_t - x^*, \nabla f_t(x_t) \rangle$$

$$= O(\sqrt{T})$$

$$R^{\text{FTRL}}(T) \leq \frac{1}{2\eta} D^2 + \sum_{t=1}^T \eta G^2$$

$$= GD\sqrt{T} \quad \text{when } \eta = \frac{D}{G}\sqrt{T}.$$

How about the convex losses?

- We still need to bound $f_t(x_t) - f_t(x_{t+1})$ on $\|x_t - x_{t+1}\|$

Let's again come back to FTRL

$$x_{t+1} = \arg \min_x F_{t+1}(x)$$

By push back lemma:

$$F_{t+1}(x_{t+1}) \leq F_{t+1}(x) - \frac{1}{2\eta} \|x - x_{t+1}\|^2 \quad \forall x$$

$$\begin{aligned} \frac{1}{2\eta} \|x_t - x_{t+1}\|^2 &\leq F_{t+1}(x_t) - F_{t+1}(x_{t+1}) \\ &= \sum_{s=1}^t f_s(x_t) + R(x_t) - \sum_{s=1}^t f_s(x_{t+1}) + R(x_{t+1}) \\ &= \sum_{s=1}^{t-1} f_s(x_t) + R(x_t) - \sum_{s=1}^{t-1} f_s(x_{t+1}) + R(x_{t+1}) \\ &\quad + f_t(x_t) - f_t(x_{t+1}) \end{aligned}$$

FTRL and OMD Algorithms

Let's consider the expert problem with linear costs and the negative entropy regularizer:

$$f_t(x) = \langle \ell_t, x \rangle, \forall t, \quad R(x) = \frac{1}{\eta} \sum_i x_i \log x_i.$$

$$x_{t+1} = \arg \min_{x \in \mathcal{P}} \sum_{s=1}^t f_s(x) + \frac{1}{\eta} \sum_{i=1}^d x_i \log x_i$$

$$= \arg \min_{x \in \mathcal{P}} \underbrace{\left\langle \sum_{s=1}^t \ell_s, x \right\rangle}_{F_{t+1}(x)} + \frac{1}{\eta} \sum_{i=1}^d x_i \log x_i$$

$$L(x, \lambda) = F_{\text{em}}(x) + \lambda \left(1 - \sum_{i=1}^d x_i \right)$$

$$\frac{\partial L(x, \lambda)}{\partial x_i} = \frac{\partial F_{\text{em}}(x)}{\partial x_i} - \lambda$$

$$= \sum_{s=1}^t l_{s,i} + \frac{1}{\eta} \frac{\partial x_i \log x_i}{\partial x_i} - \lambda$$

$$= \sum_{s=1}^t l_{s,i} + \frac{1}{\eta} (\log x_i + 1) - \lambda = 0$$

$$\Rightarrow \log x_i = \eta \lambda - 1 - \eta \sum_{s=1}^t l_{s,i}$$

$$\Rightarrow x_{t+1,i} = e^{-\eta \sum_{s=1}^t l_{s,i}} / \sum_{i=1}^d e^{-\eta \sum_{s=1}^t l_{s,i}}$$

OMD:

$$x_{t+1,i} = \frac{x_{t,i} e^{-\eta l_{t,i}}}{\sum_{i=1}^d x_{t,i} e^{-\eta l_{t,i}}}$$

FTRL and OMD Algorithms

Since FTRL and OMD both have regularization terms, any connection between these two algorithms?

- FTRL is

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x) + R(x).$$

- OMD is

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \langle x, \nabla f_t(x) \rangle + \frac{1}{\eta} B_{\psi}(x; x_t).$$

Let's consider two examples corresponding to two type of gradient algorithms:

- Online gradient descent.
- Exponentiated gradient.

FTRL and OMD Algorithms

Let's consider the linear costs and the quadratic regularizer:

$$f_t(x) = \langle \ell_t, x \rangle, \forall t, \quad R(x) = \frac{1}{2\eta} \|x\|^2.$$

FTRL and OMD Algorithms

Let's consider the expert problem with linear costs and the negative entropy regularizer:

$$f_t(x) = \langle \ell_t, x \rangle, \forall t, \quad R(x) = \frac{1}{\eta} \sum_i x_i \log x_i.$$

$$x_{t+1} = \arg \min_{x \in \mathcal{P}} \sum_{s=1}^t f_s(x) + \frac{1}{\eta} \sum_{i=1}^d x_i \log x_i$$

$$= \arg \min_{x \in \mathcal{P}} \underbrace{\left\langle \sum_{s=1}^t \ell_s, x \right\rangle}_{F_{t+1}(x)} + \frac{1}{\eta} \sum_{i=1}^d x_i \log x_i$$

FTRL and OMD Algorithms

Let's define $\theta_{t+1} = -\sum_{s=1}^t \ell_s$ and $\theta_{t+1} = \theta_t - \ell_t$.

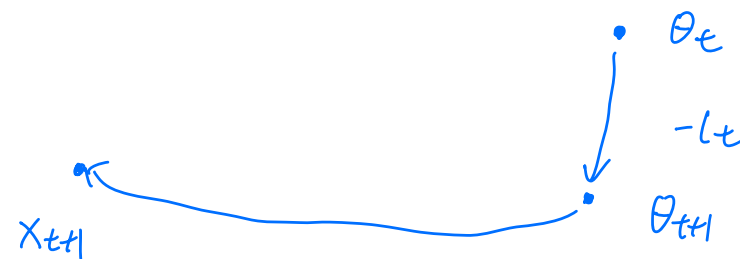
FTRL updates as

$$\theta_{t+1} = \theta_t - \ell_t$$

$$x_{t+1} = \nabla R_{t+1}^*(\theta_{t+1})$$

Primal

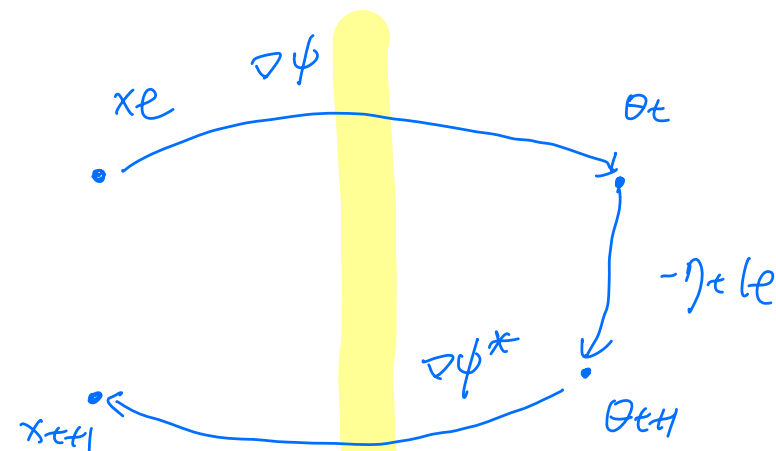
dual



Recall OMD updates as

$$\theta_{t+1} = \nabla \psi(x_t) - \eta_t \ell_t$$

$$x_{t+1} = \nabla \psi^*(\theta_{t+1})$$



FTRL v.s. OMD:

- FTRL takes “gradient” directly in dual space. Unlike in OMD, it first “mirrors” from x_t to $\theta_t = \nabla \psi(x_t)$.
- FTRL treats losses equally & OMD weights losses by η_t .

FTRL and OMD Algorithms

FTRL with the linear losses and adaptive regularization are

$$\begin{aligned}x_{t+1} &= \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x) + R_{t+1}(x) \\&= \arg \min_{x \in \mathcal{K}} \left\langle \sum_{s=1}^t \ell_s, x \right\rangle + R_{t+1}(x) \\&= \arg \max_{x \in \mathcal{K}} \left\langle - \sum_{s=1}^t \ell_s, x \right\rangle - R_{t+1}(x)\end{aligned}$$

Recall the conjugate definition $f^*(y) = \sup_x \langle y, x \rangle - f(x)$.
Therefore, we have

$$x_{t+1} = \nabla R_{t+1}^* \left(- \sum_{s=1}^t \ell_s \right)$$

FTRL and OMD Algorithms

Let's define $\theta_{t+1} = -\sum_{s=1}^t \ell_s$ and $\theta_{t+1} = \theta_t - \ell_t$.

FTRL updates as

$$\begin{aligned}\theta_{t+1} &= \theta_t - \ell_t \\ x_{t+1} &= \nabla R_{t+1}^*(\theta_{t+1})\end{aligned}$$

Recall OMD updates as

$$\begin{aligned}\theta_{t+1} &= \nabla \psi(x_t) - \eta_t \ell_t \\ x_{t+1} &= \nabla \psi^*(\theta_{t+1})\end{aligned}$$

FTRL v.s. OMD:

- FTRL takes “gradient” directly in dual space. Unlike in OMD, it first “mirrors” from x_t to $\theta_t = \nabla \psi(x_t)$.
- FTRL treats losses equally & OMD weights losses by η_t .

Follow-The-Regularized-Leader Algorithm

Follow-The-Regularized-Leader (FTRL) Algorithm

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_t(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x) + R_{t+1}(x)$.
-

We have already got the intuition on how the regularization helps stabilize the algorithm.

FTRL is a powerful framework to design online algorithms and the adaptive regularizer plays an important role.

- $R_t(x) = \sqrt{t} \|x\|^2$.
- $R_t(x) = \sqrt{t} \sum_i x_i \log x_i$.

FTRL Algorithm – Regret

Let's consider the convex costs $f_t(x)$ and the adaptive regularizer $R_t(x)$ that is “increasing” as time t and α_t -strongly convex.

Theorem 10 (convex losses and adaptive regularizer)

Assume $\|x - y\| \leq D, \forall x, y \in \mathcal{K}$ $\|\nabla f_t(x)\| \leq G, \forall x \in \mathcal{K}$.
Under Follow-The-Regularized-Leader algorithm, we have the sequence of actions $\{x_t\}$ which satisfies

$$\mathcal{R}(T) \leq R_{T+1}(x^*) - \min R_1(x) + \sum_{t=1}^T \frac{\|\nabla f_t\|^2}{2\alpha_t}.$$

We recover the good result of $O(\sqrt{T})$ (e.g., the regularizer $R_t(x) = \sqrt{t}\|x\|^2$). It is similar as FTRL with the fixed regularizer.

FTRL Algorithm – Proof

We have

$$\begin{aligned}
 \mathcal{R}(T) &= \sum_{t=1}^T f_t(x_t) - F_{T+1}(x^*) + R_{T+1}(x^*) \\
 &= \sum_{t=1}^T (F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)) \\
 &\quad + F_{T+1}(x_{T+1}) - F_1(x_1) - F_{T+1}(x^*) + R_{T+1}(x^*) \\
 &= \sum_{t=1}^T (F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)) \\
 &\quad + \underbrace{F_{T+1}(x_{T+1}) - F_{T+1}(x^*)}_{\leq 0} + \underbrace{R_{T+1}(x^*) - \min_x R_1(x)}_{=0}
 \end{aligned}$$

$f_t(x_t) + F_t(x_t) = \sum_{s=1}^t f_s(x_s) + R_t(x_t)$
 $F_{t+1}(x_{t+1}) = \sum_{s=1}^{t+1} f_s(x_{t+1}) + R_{t+1}(x_{t+1})$
 $\sum_{t=1}^T F_t(x_t) - F_{t+1}(x_{t+1}) = F_1(x_1) - F_{T+1}(x_{T+1})$

The key is to quantify $F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)$.

FTRL Algorithm – Proof

We have

$$\begin{aligned}\mathcal{R}(T) &= \sum_{t=1}^T f_t(x_t) - F_{T+1}(x^*) + R_{T+1}(x^*) \\ &= \sum_{t=1}^T (F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)) \\ &\quad + F_{T+1}(x_{T+1}) - F_1(x_1) - F_{T+1}(x^*) + R_{T+1}(x^*) \\ &= \sum_{t=1}^T (F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)) \\ &\quad + F_{T+1}(x_{T+1}) - F_{T+1}(x^*) + R_{T+1}(x^*) - \min R_1(x)\end{aligned}$$

The key is to quantify $F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t)$.

FTRL Algorithm – Proof

Lemma 11 (One-step difference)

Let F_t be α_t -strongly convex function, FTRL algorithm has

$$F_t(x_t) - F_{t+1}(x_{t+1}) + f_t(x_t) \leq \frac{\|\nabla f_t\|^2}{2\alpha_t} + R_t(x_{t+1}) - R_{t+1}(x_{t+1}).$$

FTRL Algorithm – Proof

Optimistic Follow-The-Regularized-Leader (FTRL)

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
- **Environment:** Observe the convex loss $f_t(\cdot)$.
- **Prediction:** The cost $\hat{f}_{t+1}(\cdot)$.
- **Update:**

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x) + \hat{f}_{t+1}(x) + R_{t+1}(x).$$

Intuition:

- FTRL guarantees “not too bad” even with unreliable predictions.
- Decrease the cost further if $\hat{f}_{t+1}(\cdot)$ is reliable.

Theorem 12 (Optimistic FTRL)

Assume $\|x - y\| \leq D, \forall x, y \in \mathcal{K}$ $\|\nabla f_t(x)\| \leq G, \forall x \in \mathcal{K}$.
 $R_t(x)$ that is “increasing” as time t and α_t -strongly convex.
Under Optimistic Follow-The-Regularized-Leader algorithm,
we have the sequence of actions $\{x_t\}$ which satisfies

$$\mathcal{R}(T) \leq R_{T+1}(x^*) - \min R_1(x) + \sum_{t=1}^T \frac{\|\nabla f_t - \nabla \hat{f}_t\|^2}{2\alpha_t}.$$

As in OMD with prediction, we have a few observations:

- If the predictions are “perfect”, the regret is constant!
- If the predictions are “bad”, the regret can be $O(\sqrt{T})$.
- If the predictions are “good”, the regret can be $o(\sqrt{T})$.

Optimistic FTRL – Proof

The idea is to do reduction to FTRL. Define F_t and $\hat{F}_t$

$$F_t(x) = \sum_{s=1}^{t-1} f_s(x) + R_t(x) \quad \tilde{F}_t(x) = \sum_{s=1}^{t-1} f_s(x) + \hat{f}_t(x) + R_t(x) \\ = F_t(x) + \hat{f}_t(x)$$

Recall in the proof Theorem 10, we have

$$R(T) \leq R_{T+1}(x^*) + \sum_{t=1}^T \frac{\|\nabla h_t(x)\|^2}{2\alpha_t} \\ = R_{T+1}(x^*) + \sum_{t=1}^T \frac{\|\nabla F_t(x_t) + \nabla f_t(x_t)\|^2}{2\alpha_t} \\ = R_{T+1}(x^*) + \sum_{t=1}^T \frac{\|\nabla f_t(x_t) - \nabla \hat{f}_t(x_t)\|^2}{2\alpha_t}$$

$$h_t = F_t + f_t$$

where the last equation holds because

$$x_t = \arg \min \hat{F}_t(x) \quad \nabla \hat{F}_t(x_t) = \nabla F_t(x_t) + \nabla \hat{f}_t(x_t) = 0$$

Online Learning with Delayed Feedback

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_{t-d}(\cdot)$.
 - **Update:** $x_{t+1} = \text{Alg}(f_1, f_2, \dots, f_{t-d})$.
-

A few examples:

- Subseasonal prediction: the prediction correct or not will be known in 2~6 weeks.
- Medical treatment: the treatment effective or not will be observed a few days or weeks.
- Dynamic pricing: the promotion working or not will be revealed a few days or weeks.

FTRL with Delayed Feedback

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
 - **Environment:** Observe the convex loss $f_{t-d}(\cdot)$.
 - **Update:** $x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^{t-d} f_s(x) + R_{t+1}(x)$.
-

Observations of FTRL with delayed feedback:

- Use all revealed feedback seen at time t .
- Large delay degrades the performance because of missing feedback $\sum_{s=t-d+1}^t f_s(x)$.

What is the regret of the algorithms?

Delay as Optimism in FTRL

Delay is “optimism” !!!

Delay as Optimism in FTRL

Initialization: $x_1 \in \mathcal{K}$.

For $t = 1, \dots, T$:

- **Learner:** Submit x_t .
- **Environment:** Observe the convex loss $f_t(\cdot)$.
- **Prediction:** The cost $\hat{f}_{t+1}(\cdot) = -\sum_{s=t-d+1}^t f_s(x)$.
- **Update:**

$$x_{t+1} = \arg \min_{x \in \mathcal{K}} \sum_{s=1}^t f_s(x) + \hat{f}_{t+1}(x) + R_{t+1}(x).$$

Delayed FTRL $\longrightarrow$ Optimistic FTRL.

Optimistic FTRL is a powerful framework that can handle the prediction and delay!

Theorem 13 (Delayed FTRL)

Assume $\|x - y\| \leq D, \forall x, y \in \mathcal{K}$ $\|\nabla f_t(x)\| \leq G, \forall x \in \mathcal{K}$.
 $R_t(x)$ that is “increasing” as time t and α_t -strongly convex.
Under Follow-The-Regularized-Leader algorithm, we have the sequence of actions $\{x_t\}$ which satisfies

$$\mathcal{R}(T) \leq R_{T+1}(x^*) - \min R_1(x) + \sum_{t=1}^T \frac{\|\nabla f_t - \nabla \hat{f}_t\|^2}{2\alpha_t},$$

where $\nabla \hat{f}_t = -\sum_{s=t-d+1}^t \nabla f_s$.

The effect caused by the delay:

$$\|\nabla f_t\|^2 \longrightarrow \|\nabla f_t + \sum_{s=t-d+1}^t \nabla f_s\|^2.$$

Let $\alpha_t = O(1/d\sqrt{T})$. Delayed FTRL achieves the regret of $O(d\sqrt{T})$, where the delay hurts the regret!